🚀 CampusPulse — Deployment, DevOps & Infrastructure Manual

Version: 1.0.0 (Production)

Classification: DevOps & Systems Administration Guide

Platforms Supported: Netlify, Docker / Docker Compose, Firebase App Hosting, Linux VPS

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1. Environment Variables Bible

CampusPulse requires specific environment variables for database connectivity, authentication tokens, and media uploads. Below is the complete configuration guide:

Variable Name	Required	Default / Example Value	Description & Security Guidelines
`MONGODB_URI`	YES	`mongodb+srv://user:pass@cluster.mongodb.net/campuspulse?retryWrites=true&w=majority`	Connection string to your MongoDB Atlas cluster. Keep credentials confidential.
`NEXTAUTH_URL`	YES	`https://campuspulse.netlify.app` (or `http://localhost:3000`)	Canonical root URL of your deployed application. Must match the exact production domain.
`NEXTAUTH_SECRET`	YES	Generated via `openssl rand -base64 32`	Secret key used by NextAuth to encrypt and sign JWT session tokens.
`JWT_SECRET`	YES	Generated via `openssl rand -base64 32`	Secret key for custom token signing and ticket verification logic.
`UPLOAD_DIR`	Optional	`./public/uploads`	Path to persistent local directory for banner images and media uploads.
`APP_URL`	Optional	`https://campuspulse.netlify.app`	Target remote URL used by the native

Variable Name	Required	Default / Example Value	Description & Security Guidelines
			Android Capacitor build to load the web interface.

> **TIP**
> **Generating Cryptographically Secure Secrets:**
> ```bash
> # In PowerShell or Bash:
> openssl rand -base64 32
> ```

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2. Netlify Deployment (Serverless Next.js 16)

CampusPulse is pre-configured for **100% free serverless hosting on Netlify**.

2.1 Configuration File (`netlify.toml`)

```
# netlify.toml
[build]
  command = "npm run build"

[build.environment]
  NODE_VERSION = "20"

[[headers]]
  for = "/*"
  [headers.values]
    X-Frame-Options = "DENY"
    X-XSS-Protection = "1; mode=block"
    X-Content-Type-Options = "nosniff"
    Referrer-Policy = "strict-origin-when-cross-origin"
```

2.2 Step-by-Step Netlify Deployment Steps:

1. Log in to your Netlify Dashboard.
2. Click **"Add new site" > "Import an existing project"**.
3. Connect your GitHub account and select repository: `noaman024/campuspulse`.
4. Netlify will automatically detect Next.js and the `netlify.toml` file.
5. In **"Site configuration" > "Environment variables"**, add the required keys:

- `MONGODB_URI`
- `NEXTAUTH_SECRET`

- `NEXTAUTH_URL` (Set to your Netlify site URL, e.g. `https://.netlify.app`)
- `JWT_SECRET`

1. Click **"Deploy site"**. Netlify will build and deploy the application.
2. Future commits pushed to `origin/main` will automatically trigger instant production redeployments.

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3. Docker & Docker Compose Deployment (Self-Hosted / VPS)

For universities running internal institutional servers or cloud VPS (AWS EC2, DigitalOcean, Hetzner, Linode), CampusPulse includes a production-grade multi-stage Docker build.

3.1 Multi-Stage `Dockerfile` Architecture:

- **Stage 1 (`deps`):** Uses lightweight `node:20-alpine` to install exact dependencies via `npm ci`.
- **Stage 2 (`builder`):** Compiles the Next.js production build (`npm run build`) with telemetry disabled.
- **Stage 3 (`runner`):** Minimal Alpine container with a non-privileged `nextjs` system user (`UID 1001`) for enhanced container security.

3.2 Launching with Docker Compose:

The included `docker-compose.yml` mounts a persistent Docker volume (`uploads_data`) so that uploaded photos and media are never lost when the container is restarted:

```
# 1. Create .env.local with production credentials
# 2. Build and start the container in detached mode:
docker compose up -d --build

# 3. View live container logs:
docker compose logs -f

# 4. Stop the container:
docker compose down
```

The application will be accessible at `http://:3000`.

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4. Google Cloud / Firebase App Hosting

For Google Cloud environments, CampusPulse includes native Firebase App Hosting specifications:

- `apphosting.yaml`:
- Auto-scaling: `minInstances: 0`, `maxInstances: 4`
- Resource allocation: `1 CPU`, `1024 MiB Memory`, `80 Concurrency`
- Runtime environment variable bindings.
- `firebase.json`: Configures `frameworksBackend` targeting region `us-central1`.
- `.firebaserc`: Maps to default project `campuspulse-app-f6422`.

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5. Production Health Monitoring & Maintenance

- **Log Auditing:** In Docker, run `docker logs campuspulse-app --tail 100`. In Netlify, review real-time deploy and function logs from the dashboard.
- **Database Maintenance:** Use the included database seeder (`node scripts/seed.js`) to re-sync default demo data or verify Atlas connectivity.
- **Backups:** Use MongoDB Atlas automated daily snapshots to protect student registrations and survey data.